

[Outdoor](#)[Husky](#)[Husky A300](#)[Maintenance](#)

Husky A300 Maintenance



WARNING

Always follow the proper Lock-Out Tag-Out Procedure for Husky A300 when completing inspections or maintenance.

Access to live voltage inside Husky A300 may occur if the proper Lock-Out Tag-Out procedure is not followed. This may result in electric shock.



CAUTION

Always follow safe handling procedures when performing maintenance.

Sharp edges may exist inside the robot.



CAUTION

Always follow facility guidelines to clean up fluid spills.

These may present a slip hazard.

1. Maintenance Safety

1.1 Lockout Tagout



Lockout procedures remove potential energy from machines as we perform maintenance. This reduces risk of harm while performing maintenance. The Husky's main source of potential energy is the batteries. We cannot completely discharge these batteries without damaging them, but we can disconnect them from all the robot's electronics.

- ▶ To disconnect the batteries, and lockout the machine:

SAFE MAINTENANCE PRACTICES

Follow the guidance below while performing maintenance to ensure that the system is in a safe state following any maintenance work on the robot.

1. Clean up all tools and ensure that no tools or installation materials have been left inside the robot.

2. Ensure that all cables have been tightened to the specified torque values and that there are no loose or unconnected cables.
3. Reinstall any fuse covers that were removed.
4. Restore all terminal boots to their original position prior to maintenance.

Once all parts have been reinstalled and maintenance is completed, perform the following system checks:

1. Power on the robot.
2. Place the robot up on blocks and confirm that each source of an emergency stop can be used to trigger the emergency stop condition by:
 - i. confirming that the joystick can be used to make the wheels spin
 - ii. engaging the stop condition
 - iii. confirming that the corner lights go into a flashing red light pattern
 - iv. confirming that the joystick can no longer make the wheels spin
 - v. disengaging the stop condition
 - vi. pressing the restart button
 - vii. confirming that the corner lights return to the normal driving pattern
 - viii. confirming that the joystick can be used to make the wheels spin again.

i NOTE

Perform this check of emergency stop functionality individually for each of:

- i. *Front Emergency Stop Button*
- ii. *Rear Emergency Stop Button*
- iii. *Rear Charge Port Door*
- iv. *Wireless Emergency Stop Button* (if present)
- v. *Additional Emergency Stop Buttons* (if present)

3. Connect the manual charger and charge the robot for at least 5 minutes. Confirm that the state of charge is increasing and that there are no unexpected sounds or smells.
4. With the robot in an open area at least 10 m in diameter and away from other people, connect the joystick and drive the robot for at least 30 seconds to confirm it is performing as expected.

2. Weekly Preventative Maintenance

Perform the following tasks at least once every week.

2.1 Circle Check

Perform a circle check around Husky daily to ensure no damage has occurred since its last usage. During this visual inspection, ensure the robot is turned on and verify that:

- The Status Lights are functioning correctly by pressing an Emergency Stop button and checking for the Emergency Stop light pattern.
- The robot was not damaged during the last operation.
- The cable ties used to secure all four sets of motor cables are not broken (there should be two cable ties on each motor's cables).
- All four sets of the motor cables are not touching the adjacent wheel, and are secured by cable ties in a way that would prevent this from happening.
- The motor cables have not been damaged due to contact with the wheels.

NOTE

Husky is an all-terrain IP54-rated robot, but it is not waterproof. Care should be taken that no part of the main chassis is ever submerged in water. When the chassis becomes wet or dirty, wipe it down with a damp cloth and dry with a towel.

If water is suspected to have entered the Husky chassis, power off the robot, open the top plate to ensure that any water has drained out through the drain plug, and allow Husky to dry fully (for a minimum of 24 hours) prior to putting it back into service.

2.2 Clean or Replace Air Filter

The air filter is used to prevent dirt and dust from entering the robot at the front air intake. It must be cleaned or replaced on a regular basis. High-dust environments will require more frequent maintenance.

NOTE

The air filter is foam and can be washed, then reused.

Procedure:

1. Remove the front Livery panel by loosening the quarter-turn latches and lifting the panel up. Be careful not to damage the Emergency Stop Button while removing the panel.



2. Remove the air filter.



3. Wash the filter with water and allow it to dry fully. (Alternatively, replace with a new filter.)
4. Install the clean filter.
5. Replace the front livery panel by aligning the pins and tighten the quarter-turn latches.

2.3 Battery Cell Balancing

Husky is powered by LiFePO₄ batteries. The function and health of these batteries is monitored by a Battery Management Unit—(BMU)—which is internal to each of the battery packs.

Over the operating life of Husky, the State of Charge—(SoC)—of the batteries may start being misreported, accompanied by a higher voltage difference between the batteries' cells. Ideally; battery cells should be within a range of 50 mV. If the cell voltage difference is larger than 50 mV; Husky's batteries may suddenly drop in SoC while operating, or the Husky may unexpectedly shut off.

You should cell balance the batteries at least once per week to avoid this issue.

❗ BATTERY CELL BALANCE PROCEDURE

1. Place the robot on a level surface.
2. Power off the robot.
3. Confirm the robot's circuit breaker is in the On position.
4. Connect the manual charger.
5. Let the batteries charge for at least 12 hours.

📘 NOTE

The intention of the Battery Cell Balance is to reach 100% SoC, and then charge for an additional hour while the robot is off. The suggested charge times above may be shorter if:

1. Husky's batteries are already close to 100% SoC,
2. Or you are using an upgraded manual charger.

3. Monthly Preventative Maintenance

Perform the following tasks once every month.

3.1 Tire Pressure

The Husky's tires get filled through the Schrader valve stem. This stem is close to the rim, to avoid damage from door frames and other obstacles. Some air pumps' chucks are too large to seal against this Schrader valve.

We have included a Schrader valve extension in the Husky's spare parts kit.



Husky A300 Tire's Valve Stem, With An Extension

! INFO

The Husky's tires should all be between 70 kPa to 140 kPa (*10 psi to 20 psi*).

We suggest 140 kPa for use cases on flat asphalt or concrete, to improve turning performance.

Lower tire pressures improve traction in rough and varied terrain, with rocks or other obstacles.

Lower tire pressures also reduce the effective diameter of the wheel, which will increase the error of odometry readings. Odometry is calculated using the average diameter of a tire inflated to 137 kPa (*20 psi*).

3.2 Drivetrain Inspection

It is recommended that an auditory inspection of the robot drivetrain be performed once per month. Take note of any clicking, grinding or irregular skipping sounds while the robot is moving. Note that some clicking in the gearheads is normal during stops and starts, due to backlash between the gears.

In addition, one wheel at a time, slowly roll Husky through full wheel revolution while inspecting the tire for excessive wear, cuts, or punctures.

4. Annual Preventative Maintenance

Perform the following tasks once every year.

4.1 Safety Function Checks

NOTE

No planned maintenance of the safety system is required. The checks below ensure that the safety systems remain operational.

4.1.1 Emergency Stop Buttons Check

1. Confirm that the robot can be driven normally and then stop driving.
2. Press the front emergency stop button, attempt to drive the robot again, and confirm that it does not move.
3. Clear the front emergency stop and confirm that the robot can be driven again normally.
4. Repeat the steps above for the rear emergency stop button and any wireless emergency stop buttons.

4.1.2 Debug Panel Door Check

1. Confirm that the robot can be driven normally and then stop driving.
2. Open the rear debug panel door, attempt to drive the robot again, and confirm that it does not move.
3. Close the rear debug panel door, clear the emergency stop condition, and confirm that the robot can be driven again normally.

4.1.3 Emergency Stop Braking Check

1. Remove any payload from your robot.
2. Turn the robot on.

3. Find a 10 metre stretch of flat ground. At the 5 metre point, mark the ground with a braking test starting point.
4. Drive the robot at 1 m/s.
5. Press an Emergency Stop button as the robot's front tire touches the braking test mark.
6. Measure the braking distance from the mark, to the stopped position of the front tire. Confirm that the stopping distance was less than 40 cm.

Contact our [Support Team](#) if the stopping distance was more than 40 cm.

5. 36-Month Preventative Maintenance

Perform the following tasks once every thirty-six months.

5.1 Battery Health Check

Husky's batteries are expected to last for at least 36 months of continuous operation. The actual lifetime experienced will vary greatly depending on the usage and operating environment. Contact [Support](#) for an in-depth assessment of the battery's health if you have any concerns.

6. Hardware Maintenance

6.1 Common Replacement Items

NOTE

This does not cover the full bill-of-materials for your robot. These items are things we consider critical, so you may want to keep spares on hand. You can get any replacement part by contacting support@clearpathrobotics.com, including parts not listed here.

Description	What Subsystem Is This Used In?	CPR Item	Manufacturer	Manufacturer Item	Maintenance Procedure
Fuse, 3A	<i>System Interface Assembly (F2, F3, F4, F5, F8)</i>	-	Littelfuse	0453003.MR	Replacing board mount fuses
Fuse, 5A	<i>System Interface Assembly (F6, F10)</i>	-	Littelfuse	0454005.MR	Replacing board mount fuses
Fuse, 8A	<i>System Interface Assembly (F7, F9)</i>	-	Littelfuse	0453008.MR	Replacing board mount fuses
Fuse, 10 A	<i>System Interface Assembly (F52, F53, F54, F55, F56, F57, F58)</i>	-	Littelfuse	0297010.WXT	Replacing blade mini fuses
Fuse, 10 A	<i>System Interface Assembly, and System Interface Assembly's Standby Power Cable</i>	018897	Littelfuse	0297010.WXNV	Replacing blade mini fuses

Description	What Subsystem Is This Used In?	CPR Item	Manufacturer	Manufacturer Item	Maintenance Procedure
Fuse, 20 A	<i>System Interface Assembly</i>	032295	Littelfuse	0297020.U	Replacing blade mini fuses
Fuse, 40 A	<i>Power Distribution's Large VBAT Contactors</i>	023567	Littelfuse	142.5631.5402	Replacing BF1 fuses
Fuse, 50 A	<i>Power Distribution's Manual Charger</i>	029139	Littelfuse	142.5631.5502	Replacing BF1 fuses
Fuse, 60 A	<i>Batteries, Power Distribution's Mobility Contactors, Power Distribution's Shunt Brake</i>	029140	Littelfuse	142.5631.5602	Replacing BF1 fuses
Fuse, 150 A	<i>Batteries</i>	029145	Littelfuse	142.5631.6152	Replacing BF1 fuses
Air Filter	<i>Fans</i>	032403	Clearpath Robotics	-	Air Filter Maintenance
Tire	<i>Mobility</i>	000269	Carlisle	5100201	Tire Maintenance

Description	What Subsystem Is This Used In?	CPR Item	Manufacturer	Manufacturer Item	Maintenance Procedure
Tire's Inner Tube	<i>Mobility</i>	000435	Carlisle	323790	Tire Maintenance

6.2 Critical Items

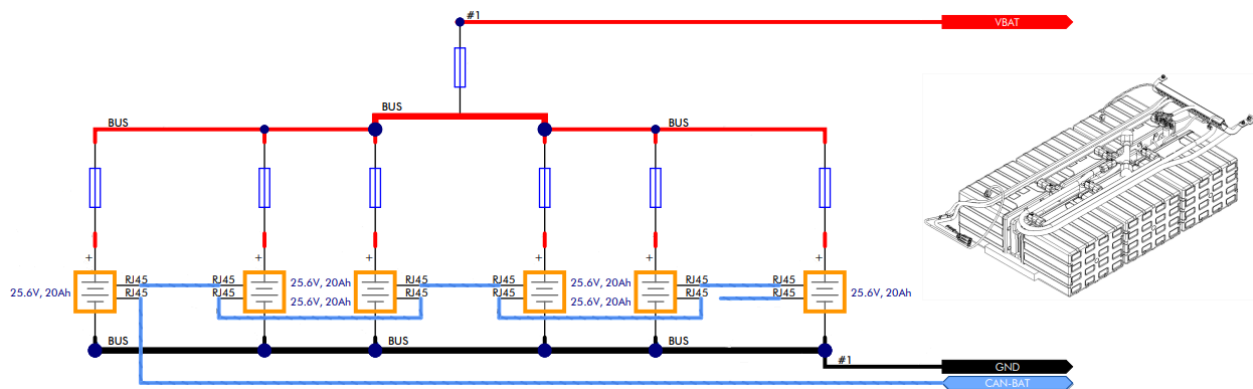
! INFO

These items are unlikely to fail in normal use. We are providing this data because some customers want spares on hand for anything that could impact their schedule.

Description	What Subsystem Is This Used In?	CPR Item	Manufacturer
Battery	<i>Batteries</i>	025047	Clearpath Robotics
Battery Manual Charger	<i>Spare Parts</i>	032729	Clearpath Robotics
Main PCBA	<i>System Interface Assembly</i>	030449	Clearpath Robotics
BLDC Motor Controller PCBA	<i>System Interface Assembly</i>	030377	Clearpath Robotics
Common Core PCBA, (MCU)	<i>System Interface Assembly</i>	030446	Clearpath Robotics
12 V DC-DC PCBA	<i>System Interface Assembly</i>	030430	Clearpath Robotics

Description	What Subsystem Is This Used In?	CPR Item	Manufacturer
24 V DC-DC PCBA	<i>System Interface Assembly</i>	030433	Clearpath Robotics
Gearmotor	<i>Mobility</i>	031514	Clearpath Robotics
Fan	<i>Fans</i>	031356	Clearpath Robotics
Computer	<i>Computer</i>	032741, or 032742	Clearpath Robotics
Network Switch	<i>Network Switch</i>	033443	Clearpath Robotics

6.3 Wiring Diagram



Snippet Of The Husky's Wiring Diagram, Showing The Optional 120 Ah Battery

Download The Robot's Wiring Diagram

Wiring diagrams help diagnose the cause of issues, such as damaged cables or loose connections. This wiring diagram lists Clearpath's 6 digit item numbers for circuit boards and cables. The diagram does not include pinouts for individual cables.

NOTE

Please contact our [Support Team](#) if you need more debugging information, or want a replacement cable.

6.4 Tire and Inner Tube

**CAUTION**

Place the robot up on blocks prior to replacing the tire or inner tube. The robot may tip over if you remove a wheel without stabilizing the chassis.

- ▶ To replace the tire and inner tube:

6.5 Motors



Husky's Gearmotor, With Power And Data Cables

- ▶ To remove the motors:

6.6 Front Cover



The Husky's identification labels, status lights, air filter, and some fasteners are located under the *Front Cover*.

► To remove this cover:

6.7 Nameplate And Regulatory

Husky's *Nameplate* is located under the *Front Cover*. The *Nameplate* includes regulatory marks, the product's name, serial number, and date of manufacture.

There will be other labels on the chassis with FCC information, such as the Intel Wi-Fi card shown in the attached image.



Husky's Nameplate Under The Front Cover

TIP

Our [Removing The Front Cover](#) section explains how to access the *Nameplate*.

6.8 Removing The Top Plate

To remove Husky's *Top Plate*:

1. Turn off the Husky.
2. Open the *Rear Charge Port Door*.
3. Flip the *Circuit Breaker* to the Off position.
4. Close the *Rear Charge Port Door*.
5. Remove the seven M6×1 socket head screws from the *Top Plate*. Note that one of the screws also has a sealing washer attached to it.
6. If applicable, remove any of your custom cables from the *Top Plate*'s connector openings.
7. Lift the *Top Plate* off of the robot.

NOTE

When reinstalling the *Top Plate*:

- Torque the tray's 4 screws to 3 N·m.

DANGER

Do not drag the *Top Plate* on the robot, because it may damage the bulb seal around the Husky's chassis.

Top Plate's Seventh Screw, With A Sealing Washer

6.9 Electronics Tray

Top Plate Removed, With The Husky's Front Surface On The Right Side Of The Image

Once you have opened Husky's top, the first thing you see is the *Electronics Tray*. This tray includes the:

- Husky's *System Interface Assembly* circuit board
- Primary computer
- IMU
- Optional secondary computer or graphics card, (*not shown in the image*)
- Optional USB hub, (*not shown in the image*)

You may want to remove the *Electronics Tray* to access the integration bay under the tray, or to maintain Husky's *Batteries* and *Power Distribution* components.

6.9.1 Hinging The Electronics Tray

Tray Hinged, And Then Held With A Prop Rod

- ▶ To hinge the tray:

EXPOSED +V BATTERIES

Lifting the tray will expose Husky's batteries. The batteries's exposed +V terminals are always live, even when Husky's circuit breaker is in the Off position. The batteries are fused, but can still produce an electric arc if shorted to ground.

Wear protective equipment to guard your skin and eyes.

Keep the workspace clean, without any combustible materials nearby.

Review your workplace's safety procedures related to fires of LiFePO₄ batteries.

6.9.2 Removing The Electronics Tray

Tray Removed, But Still Tethered By The USB And HDMI Cables

▶ To remove the tray:

6.10 Battery Integration Tray

The Husky configurations with 40 Ah and 80 Ah batteries include a metal tray inside the chassis. You can mount your custom integration components on this tray.

NOTE

The 120 Ah Husky fills the entire chassis volume with batteries.

▶ To remove the tray:

6.11 Batteries

The Husky's batteries are visible once you remove the *Electronics Tray*. The batteries +V terminals are always live, even when the circuit breaker is in the Off position. You may want to remove the

batteries to replace them, or to access the *Power Distribution* subassembly.

NOTE

The images show the Husky 40 Ah configuration which has 2 batteries. The 80 Ah and 120 Ah configurations follow a similar process, but with 4 or 6 batteries.

INFO

Husky's batteries should last 3000 charge cycles if they are charge balanced at least once per week, and not left in a depleted state.

If you order new batteries from Clearpath Robotics, ask us to configure the batteries for a Husky A300. We will configure the battery software addresses and CAN address before shipping them to you.

▶ To remove the batteries:

6.12 Rear Cover

▶ To remove the *Rear Cover*

6.13 Bumpers

Bottom Of A Husky A300

▶ To remove the bumpers:

6.14 Air Grilles

Husky's Front Air Grille, Behind The Bumper

The front and rear *Air Grilles* prevent large debris from entering the robot.

- ▶ To remove the grilles

6.15 Fans

Husky has 4 fans at the front of the robot. These fans operate similar to a server; as all 4 push air into the robot to create positive pressure inside the chassis.

- ▶ To remove the fans:

6.16 Power Distribution

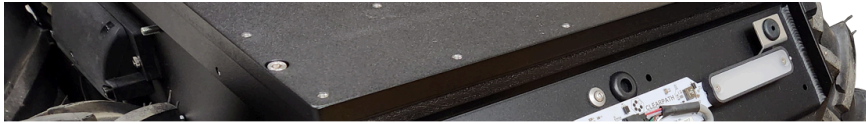
NOTE

The *Power Distribution* subassembly includes all the robot's large power electronics:

- Main relay contactor
- Mobility relay contactors
- Shunt brake relay contactors
- Shunt brake resistor
- Large VBAT contactors
- Current sensor
- Circuit breaker
- Fuses
- Busbars, and cables to the batteries

- ▶ To remove the *Power Distribution* subassembly:

6.17 Status Lights



View Inside The Husky's Chassis

- ▶ To remove the light assemblies:

- ▶ To remove the PCBA from the assembly:

6.18 Network Switch

The network switch connects the primary computer, the rear debugging port, and any additional Ethernet devices.

- ▶ To remove the switch:

- ▶ To remove the switch from its subassembly brackets:

6.19 Computer

Husky's Primary Computer, On The Left/Port Side Of The Robot

There are 2 computer mounting locations on Husky's *Electronics Tray*. Both mounting locations are sized for Mini-ITX computers.

**TIP**

Clearpath offers a standard computer with an Intel i3 processor, or a performance option with a i7 processor—though custom builds are also an option.

► To remove a computer:

6.20 Replacing CMOS Battery

ASRock IMB-1232-WV motherboard for an Intel 13th gen processor

**INFO**

These instructions were developed in 2026 for our robots' mini-ITX computer with a 13th gen intel processor.

Our computers include a CMOS battery to maintain configuration settings while the robot is off. The CMOS battery will recharge whenever the robot is running. The battery typically needs to be replaced after 1 - 2 years.

**NOTE**

A common symptom of a depleted CMOS battery is the computer not turning on. Your robot's microcontroller may turn on, so lights and fans may run; but the robot will not drive.

Motherboards's connector for the CMOS battery

► To change the battery:

**NOTE**

You will need to update the computer's BIOS settings after replacing the battery.

6.21 Debugging The System Interface Assembly, (*Debug LEDs*)

The System Interface Assembly includes many LEDs to help debug system issues.



WARNING

The Husky must be powered on, with the Top Plate removed to see these Debug LEDs. This exposes significant battery energy to the technician, and risk of starting a fire.

Review your worksite's safe operating procedures before performing this task.

▶ [Click to see all the LED functions](#)

6.22 Replacing Circuit Boards

Husky's *System Interface Assembly*, including DC-DCs, BLDC controllers, Common Core, and fuses

Husky has several circuit boards—(*PCBAs*). It is uncommon for a PCBA to fail during operation, but could occur from physical damage during a maintenance task. If a failure occurs, usually only one board needs to be replaced, rather than the full *System Interface Assembly*.



ESD PROTECTION

PCBAs are sensitive components and can be damaged easily through electrostatic discharge—(*ESD*). Be sure to follow proper ESD protection measures prior to handling any circuit boards.

WARNING



Confirm that Husky's Emergency Stop System is functioning after changing any of the circuit boards.

▶ Procedure for replacing the DC-DC and BLDC circuit boards

▶ Procedure for replacing the Common Core circuit board

▶ Procedure for replacing the Main circuit board

6.23 Replacing Board Mount Fuses

1. Perform the [Lockout-Tagout Procedure](#).
2. Locate the fuse to be replaced. Follow the maintenance procedures for the corresponding subassembly to access the fuse.



CAUTION

Treat all circuits, cables, and fuses inside the robot as potentially live.

Some connections inside the robot are live even after Lockout Tagout. Refer to the details of the subassembly maintenance procedures to determine if the fuse is connected to a live connection.

3. Use tweezers to pull up on the old fuse to remove it from the fuse holder.
4. Slide the replacement fuse into the fuse holder and push firmly until the fuse is installed fully.

NOTE

Ensure that you are replacing the fuse with the same part number and rating. Refer to [Common Replacement Items](#) for a list of fuses.

6.24 Replacing Mini Blade Fuses

1. Perform the [Lockout-Tagout Procedure](#).
2. Locate the fuse to be replaced. Follow the maintenance procedures for the corresponding subassembly to access the fuse.



CAUTION

Treat all circuits, cables, and fuses inside the robot as potentially live.

Some connections inside the robot are live even after Lockout Tagout. Refer to the details of the subassembly maintenance procedures to determine if the fuse is connected to a live connection.

3. Pull up on the old fuse to remove it from the fuse holder.
4. Slide the replacement fuse into the fuse holder and push firmly until the fuse is installed fully.

NOTE

Ensure that you are replacing the fuse with the same part number and rating. Refer to [Common Replacement Items](#) for a list of fuses.

6.25 Replacing BF1 Fuses

1. Perform the [Lockout-Tagout Procedure](#).
2. Locate the fuse to be replaced. Follow the maintenance procedures for the corresponding subassembly to access the fuse.



CAUTION

Treat all circuits, cables, and fuses inside the robot as potentially live.

Some connections inside the robot are live even after Lockout Tagout. Refer to the details of the subassembly maintenance procedures to determine if the fuse is connected to a live connection.

3. Remove the fuse cover.
4. Remove the nuts holding the fuse in place.

5. Remove the fuse from the fuse holder.
6. Install the replacement fuse into the fuse holder.

NOTE

Ensure that you are replacing the fuse with the same part number and rating. Refer to [Common Replacement Items](#) for a list of fuses.

7. Reinstall both nuts onto the fuse holder and tighten to 3 N·m unless otherwise specified in the subassembly maintenance procedures.
8. Reinstall the fuse cover.

7. Software Maintenance

7.1 Getting New Packages

NOTE

If you are upgrading your robot from an older version of ROS, please refer to our upgrade instructions for upgrading to [ROS 1 Melodic](#), [ROS 1 Noetic](#) and [ROS 2](#).

Clearpath Robotics robots are always being improved, both its own software and the many community ROS packages upon which it depends! You can use the `apt` package management system to receive new versions all software running on the platform.

TIP

Use the `apt` command for package management commands in this guide.

Each robot leaves the factory already configured to pull packages from <http://packages.ros.org> as well as <http://packages.clearpathrobotics.com>. To update your package and download new package versions, simply run:

```
sudo apt update
sudo apt full-upgrade
```

7.2 MCU Firmware Update

Refer to [firmware update instructions](#). Be sure to select the ROS 2 distribution that matches the ROS 2 distribution installed on your robot.

7.3 Motor Controller Firmware Update

Use the following command to update the firmware on the motor controllers, using the serial number of your robot rather than `xxxxx`.

CAUTION

Do NOT power off the robot while this command is running. The update typically takes a few minutes to complete for all four motor controllers.

```
ros2 action send_goal /a300_xxxxx/platform/motors/update  
clearpath_motor_msgs/action/LynxUpdate {} -f
```

7.3.1 What to Expect

The update is implemented as a [ROS 2 action](#), so the command will stream feedback while it runs and then print a final result. Both pieces of output are normal and should be reviewed to confirm a successful update.

Feedback (during the update)

While the update is in progress, the action publishes a `Feedback` message containing a `progress` array, with one entry per motor controller (front-left, front-right, rear-left, rear-right). Each value is the percentage of the firmware image that has been transferred to that controller (0.0 - 100.0). A typical feedback block looks like:

```
Feedback:  
  progress:  
    - 100.0  
    - 100.0  
    - 100.0  
    - 99.0
```

The values increment as each controller is flashed and will end at `100.0` for every controller that updated successfully. It is normal to see one controller briefly lag behind the others.

Result (when the update finishes)

When the action completes, a `Result` message is printed with a `success` array, again with one entry per motor controller, in the same order as the feedback `progress` array. A `true` value means that controller was re-flashed with the new firmware; a `false` value means that controller was already running the target firmware version (or did not need to be updated) and was therefore skipped. The action itself reports `Goal finished with status: SUCCEEDED` in both cases; `false` entries are **not** an error.

```
Result:
  success:
    - false
    - false
    - false
    - false
```

```
Goal finished with status: SUCCEEDED
```

The example above is what you will see if all four motor controllers are already on the latest firmware: the action succeeds, but no controller actually needed to be reflashed.

NOTE

If the action exits with a status other than `SUCCEEDED`, or if any feedback `progress` entry stops increasing and never reaches `100.0`, contact [Clearpath Support](#).

Once the action has completed, power cycle the robot for the new firmware to take effect.

7.4 Motor Controller Calibration

The motor controllers are configured in the factory and do not normally need to be reconfigured in the field. If you have been instructed by Clearpath Support to recalibrate, place the robot up on blocks so that the wheels do not touch the ground, and then use the command below, using the serial number of your robot rather than `xxxxx`.

 CAUTION

Do NOT power off the robot while this command is running.

 INFO

Ensure that the robot does not have its Emergency Stop engage while running this command.

 INFO

Ensure that the robot is up on blocks so that the wheels do not touch the ground and can spin freely.

```
ros2 action send_goal /a300_XXXXX/platform/motors/calibrate  
clearpath_motor_msgs/action/LynxCalibrate {} -f
```

8. Support

Clearpath is committed to your success. Please get in touch with us and we will do our best to get you rolling again quickly: support@clearpathrobotics.com.

To get in touch with a salesperson regarding Clearpath Robotics products, please email research-sales@clearpathrobotics.com.

If you have an issue that is specifically about ROS and is something which may be of interest to the broader community, consider asking it on [Robotics Stack Exchange](#). If you do not get a satisfactory response, please ping us and include a link to your question as posted there. If appropriate, we will answer on Robotics Stack Exchange for the benefit of the community.

*Last updated on **May 13, 2026***